

Supporting Information for:
Bayesian Multilevel Modeling of Retention Data
Informed by Structural Similarity of Analytes

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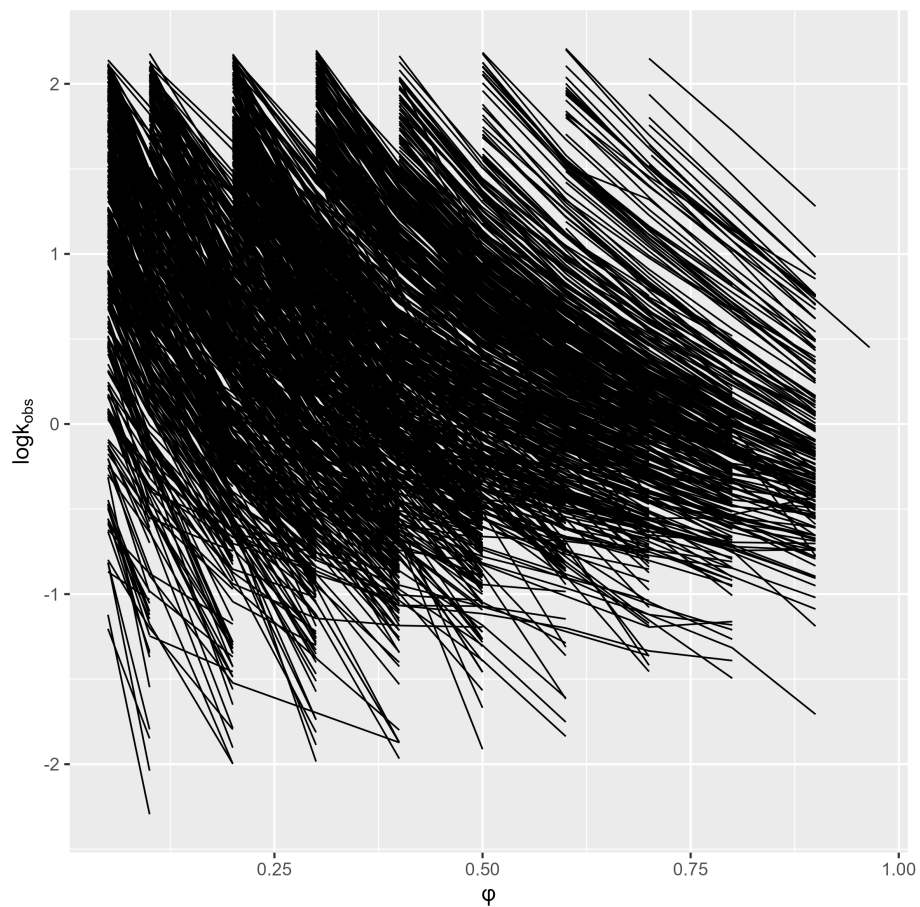
1 Table S1. Summary of the MCMC simulations of the marginal posterior distributions of population-level parameters

variable	mean	median	sd	mad	q5	q95	rhat
θ_{logkw}	3.05	3.05	0.19	0.19	2.74	3.36	1.00
θ_{S1}	3.67	3.67	0.16	0.16	3.40	3.94	1.00
θ_{dlogkw}	-0.35	-0.35	0.12	0.12	-0.55	-0.15	1.00
θ_{dS1}	0.30	0.30	0.09	0.09	0.16	0.45	1.00
β_{logkw}	0.86	0.86	0.05	0.06	0.77	0.95	1.00
β_{S1}	0.62	0.62	0.05	0.05	0.54	0.70	1.00
θ_{S2}	2.90	2.90	0.03	0.03	2.85	2.95	1.02
ω_{logkw}	1.41	1.41	0.04	0.04	1.35	1.47	1.00
ω_{S1}	1.13	1.13	0.03	0.03	1.08	1.18	1.00
$\rho_{1,2}$	0.88	0.88	0.01	0.01	0.87	0.90	1.00
σ	0.05	0.05	0.00	0.00	0.05	0.05	1.00
α	0.71	0.71	0.04	0.04	0.64	0.77	1.00
$\kappa1_{logkw}$	0.15	0.14	0.02	0.02	0.11	0.18	1.01
$\kappa1_{S1}$	0.13	0.13	0.02	0.02	0.10	0.17	1.01
$\kappa2_{logkw}$	0.34	0.33	0.06	0.06	0.23	0.44	1.00
$\kappa2_{S1}$	0.19	0.18	0.07	0.07	0.08	0.30	1.01

mean denotes the posterior mean; median denotes the posterior median, sd denotes the posterior standard deviation; mad denotes the median absolute deviation; q5 denotes the 5th percentile of the posterior draws; q95 denotes the 95th percentile of the posterior draws; rhat denotes the Gelman–Rubin convergence diagnostic. Values close to 1.0 indicate that the MCMC chains have mixed and converged well.

2 Figure S1. Raw data

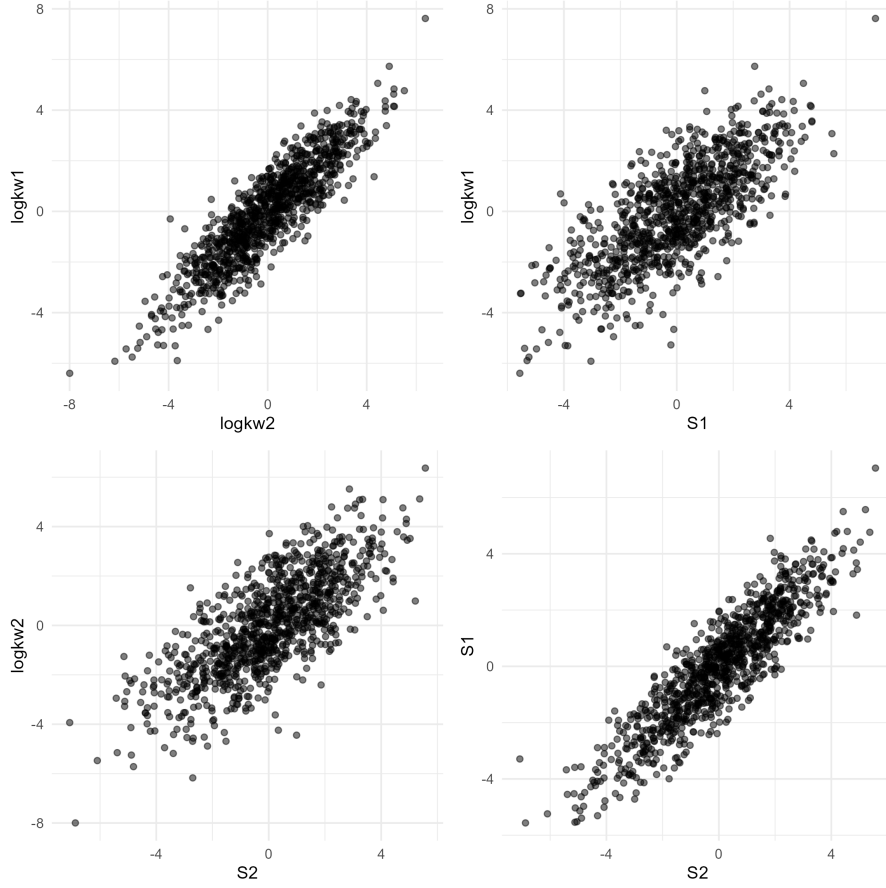
The relationship between observed logk and organic modifier content for all the analytes.



3 Figure S2. Matrix Normal Distribution

Simulations from the Matrix Normal Distribution with row correlations of 0.9 and column correlation of 0.75.

$$\begin{bmatrix} \log k_{w1} & S_1 \\ \log k_{w2} & S_2 \end{bmatrix} \sim \mathcal{MN} \left(\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.9 \\ 0.9 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0.75 \\ 0.75 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \right)$$



4 Figure S3. Functional group effects

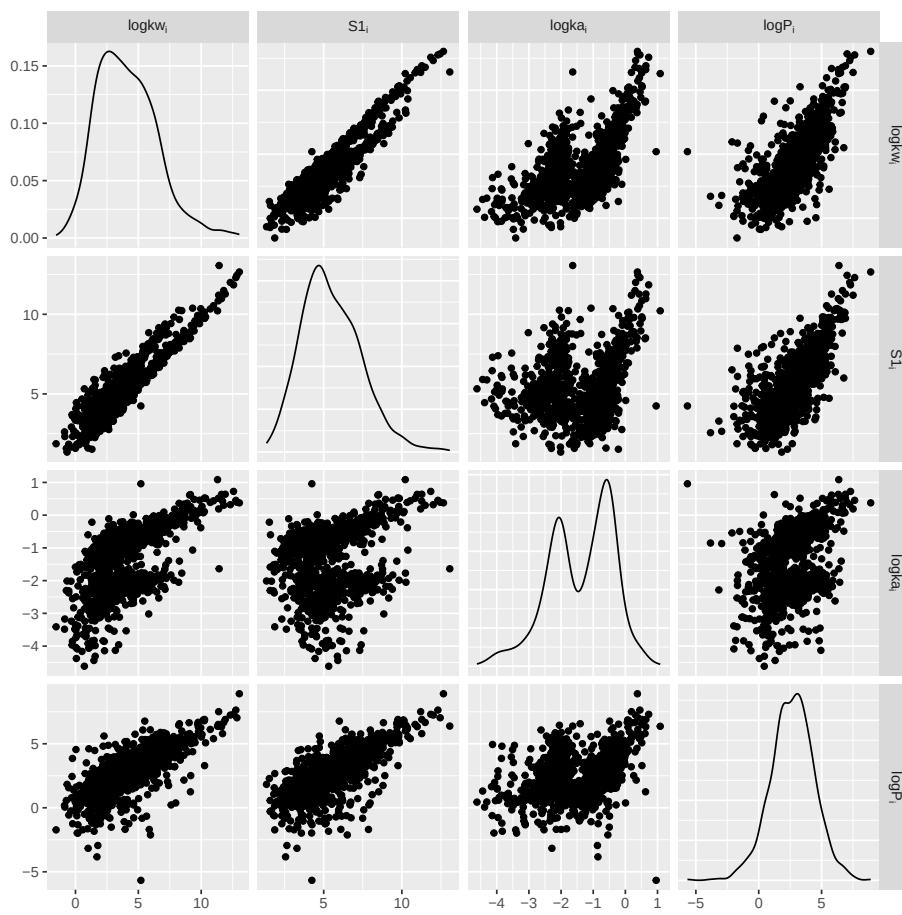
A summary of the marginal posterior distributions of the functional group effects. Red represents the effect of dissociated functional groups, and green represents the effect of non-dissociated functional groups. The least common functional groups (≤ 10 across all analytes) are presented. The plot for the remaining functional groups is presented in Figure 2.



5 Figure S4. Individual parameters, R_i

$\log k_a$ is the (extrapolated) log retention in 100% ACN:

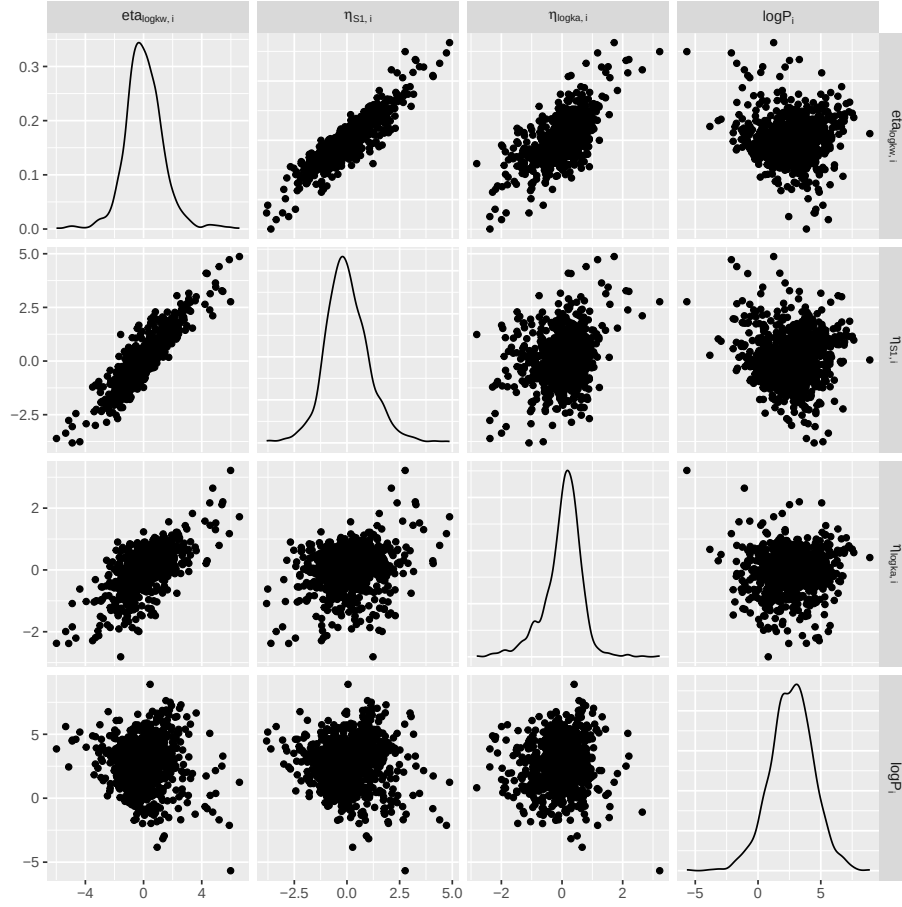
$$\log k_{a_i} = \log k_{w_i} - S_{1,i}$$



6 Figure S5. Individual parameters, $\eta_{R,i}$

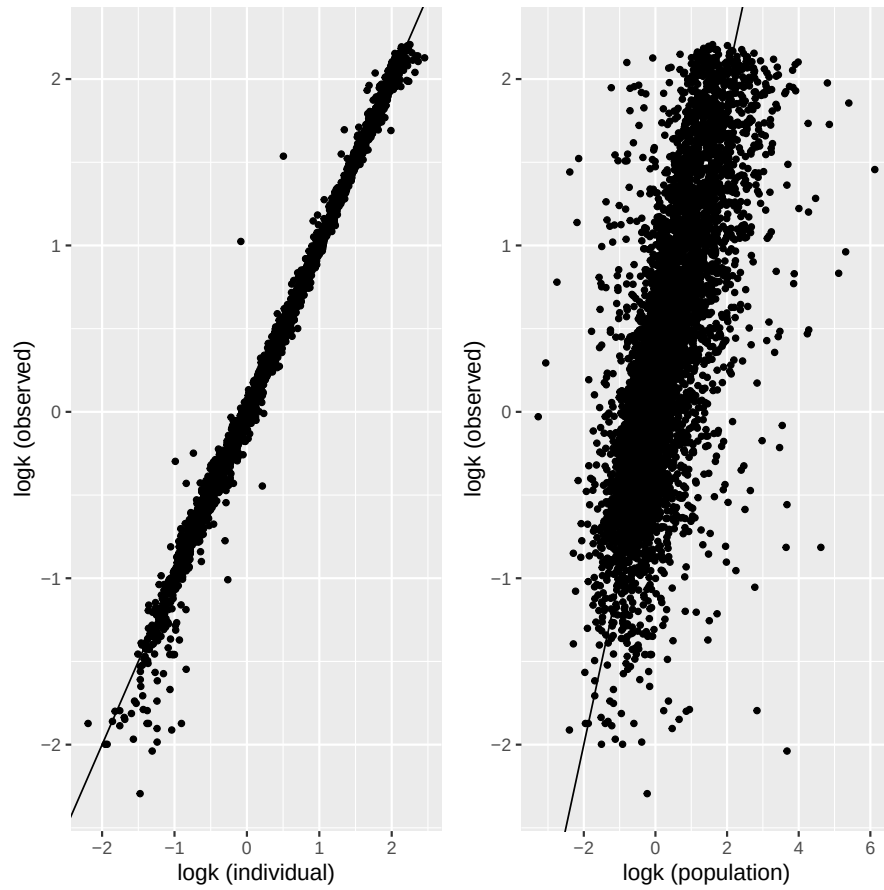
$\eta_{R,i}$ represents the deviation of an individual analyte's parameter from its expected (population-level) value.

$$\eta_{R,i} = R_i - (\theta_R + \beta_R \cdot \log P_i + \pi_{R1} \cdot X_{1i} + \pi_{R2} \cdot X_{2i})$$



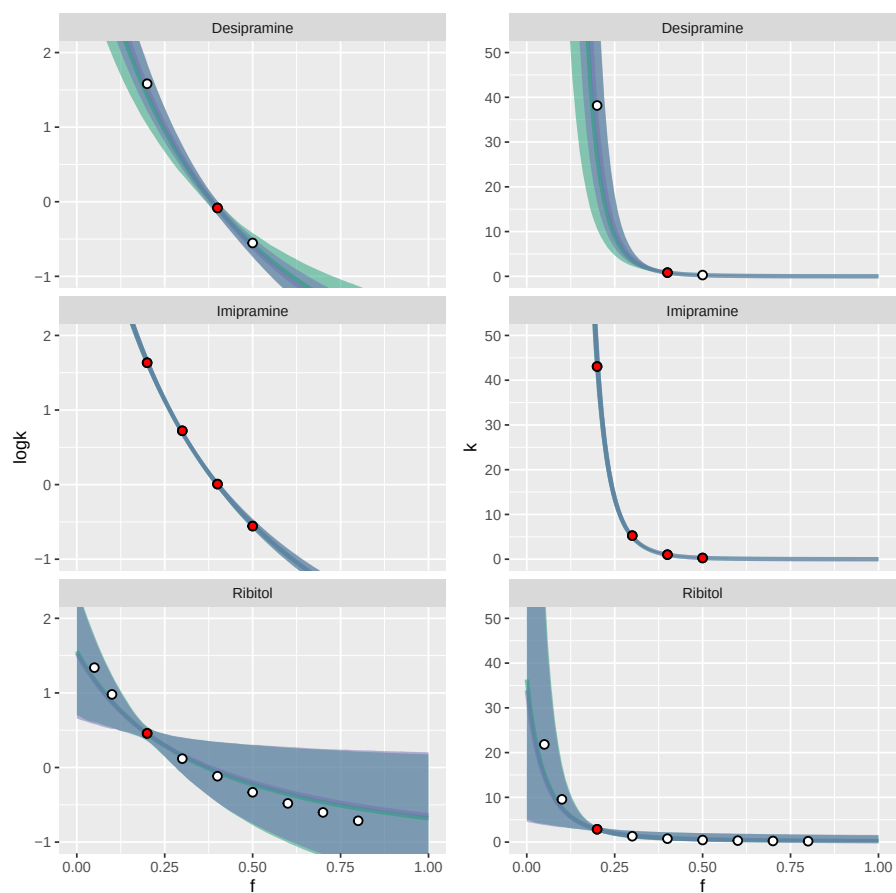
7 Figure S6. Goodness of fit plots (GOF)

Observed vs. population and individual predictions:



8 Figure S7. Example illustrating model application

The green area represents individual prediction given the model and experimental data shown in red and assuming no structural similarity. The purple area includes similarity matrix in predictions. The available observations are presented as closed white symbols. Both the logk and k are shown for comparison.



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Original Computing Environment

R version 4.3.1 (2023-06-16 ucrt)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64 (build 19045)

Matrix products: default

locale:

```
[1] LC_COLLATE=Polish_Poland.utf8 LC_CTYPE=Polish_Poland.utf8
[3] LC_MONETARY=Polish_Poland.utf8 LC_NUMERIC=C
[5] LC_TIME=Polish_Poland.utf8
```

time zone: Europe/Warsaw

tzcode source: internal

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] cmdstanr_0.8.0.9000 kableExtra_1.4.0    dplyr_1.1.4
```

loaded via a namespace (and not attached):

```
[1] jsonlite_1.8.8      compiler_4.3.1      tidyselect_1.2.1
[4] xml2_1.3.6          stringr_1.5.1       systemfonts_1.1.0
[7] scales_1.3.0        textshaping_0.4.0   yaml_2.3.8
[10] fastmap_1.2.0       R6_2.5.1            generics_0.1.3
[13] distributional_0.4.0 knitr_1.46          backports_1.5.0
[16] checkmate_2.3.1     tibble_3.2.1        munsell_0.5.1
[19] svglite_2.1.3       pillar_1.9.0        posterior_1.6.0
[22] rlang_1.1.3         utf8_1.2.4          stringi_1.8.4
[25] xfun_0.44           viridisLite_0.4.2   cli_3.6.2
[28] withr_3.0.2         magrittr_2.0.3      ps_1.7.6
[31] processx_3.8.4      digest_0.6.35       rstudioapi_0.16.0
[34] lifecycle_1.0.4     vctrs_0.6.5         data.table_1.15.4
[37] tensorA_0.36.2.1    evaluate_1.0.3      glue_1.7.0
[40] codetools_0.2-19    ragg_1.3.2          abind_1.4-8
[43] fansi_1.0.6         colorspace_2.1-0    rmarkdown_2.27
[46] matrixStats_1.3.0   tools_4.3.1         pkgconfig_2.0.3
```

[49] `htmltools_0.5.8.1`